RealQM Molecule Simulator: Small Molecules from H, C, N, O, Validated by Forces at Observed Geometries

Claes Johnson[1]

[1]KTH Royal Institute of Technology, Stockholm, Sweden , claesjohnson@gmail.com

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Abstract

We describe a simulator for small molecules built from hydrogen, carbon, nitrogen and oxygen in RealQM, a parameter-free real-space model in which each electron is a unit-charge component on a non-overlapping spatial domain, and the validation protocol built into it. A model whose dynamics is defined by a force law should be validated on forces. We propose a protocol for this: evaluate the nuclear forces at a molecule's *measured* geometry, where the true force vanishes by definition, so that every residual is model error read directly—without a reference calculation, without an energy scan, and without locating a minimum. One calculation per molecule replaces a scan over hundreds, and the residual converts to a bond error through $\Delta R \approx F/k$ with the measured force constant. We apply this across twenty configurations built from H, C, N and O. Three internal checks ride on every calculation, and we show that which of them carries information is decided by the molecule's symmetry group: the kernel–kernel energy is exact arithmetic on the geometry built; the vanishing of the total nuclear force is *not* an identity but a convergence criterion, since it equals the net force of the electron cloud on the nuclei; and where symmetry cancels that sum, a per-atom zero at a centre of symmetry supplies a check that cancellation cannot fake. The protocol first exposed a defect in the force itself—a sphere-averaged potential gradient whose *sign* was set by the averaging radius—which we replace by the analytic derivative of the discretised electron–kernel energy, a volume integral with no free radius. It then isolates the dominant error: holding geometry and symmetry fixed and varying only the central kernel charge changes the convergence error by a factor of sixty in methane and sixteen in hydrogen cyanide, while hydrogen content, box size and off-axis placement are excluded as causes by direct test. A secondary effect is anisotropy of the cubic grid, which cannot represent threefold or bent geometries evenly. We report measured verdicts on the model's valence conventions, and note that several previously recorded failures of three-centre systems are consistent with artefacts of an unconverged energy scan rather than properties of the model. The protocol's practical virtue is that it degrades visibly: configurations that fail announce it through their own checks instead of returning plausible wrong geometries.

1 The simulator

The simulator presents a list of molecules, each stored as experimental internal coordinates converted to Cartesian form and centred on the centroid. Selecting one builds the system, sizes the

computational box from the molecule's own extent plus room for the electron clouds, chooses the grid for a target spacing, and runs. Two solvers are available: a GPU code carrying one domain per entry, capped at sixteen domains, and a second implementing the same model as one globally partitioned field, admitting two thousand (Section 4).

Three choices are built in as defaults rather than offered as options, each because it was measured rather than assumed. Valence charges are $Z_x = 1, 4, 3, 3$ for H, C, N, O—oxygen at +3 rather than the common +2, which converges five times worse at the same bond error, and carbon never at +2, which has no defence (Section 9). The nuclear force is the volume integral of Section 3.1 rather than a sampled potential gradient. And the box follows the molecule, after several early calculations were found to have been silently clipped by an undersized one.

What distinguishes the simulator from a calculator is that it displays its own error checks continuously alongside the result: the kernel–kernel energy against exact arithmetic on the geometry actually built, and the total nuclear force as a percentage of the largest single force. The reasoning behind those two quantities, and why which of them carries information depends on the molecule's symmetry, is the subject of Section 6. Their practical effect is that a calculation which cannot be trusted says so on the same screen as its answer.

2 Introduction

RealQM [1] replaces the many-electron wavefunction by N unit charge densities occupying non-overlapping spatial domains with free interfaces, interacting by Coulomb's law with each pair counted once. Nuclei enter as kernels; in molecular work a filled core is replaced by a *pseudo-kernel* of reduced charge Z_x carrying only the bonding valence. The model is therefore a force law: kernels move under the electrostatic force of the charge distribution, and a predicted geometry is wherever that force vanishes.

Validating such a model is awkward by the usual routes. Scanning the energy over geometry requires every point of the scan to be converged, and a scan assembled from partially relaxed states returns a smooth but meaningless curve. Relaxing under the force requires the force to be trustworthy, which is the thing under test. Both routes also need many calculations per molecule, and both return an answer whether or not the underlying states deserve one.

We propose instead to evaluate forces at the *observed* geometry. The logic is elementary: at the measured equilibrium the true force on every nucleus is zero, so the computed residual is the model's error, with experiment supplying the reference. No minimum is sought, no scan is assembled, and a single calculation suffices per molecule. The residual is not merely a diagnostic but a quantitative prediction, since

$$\Delta R \;\approx\; F/k \tag{1}$$

with k the measured force constant converts it to a bond-length error. For H_2, $k = 5.16\,\mathrm{mdyn/\AA} = 0.3314\,E_h/a_0^2$, so $F = 0.01\,E_h/a_0$ corresponds to a bond error of 3%.

The protocol suits a force-law model particularly well, because it tests the model on its own terms: the force is primary, and agreement with $-dE/dR$ is a consistency check rather than the definition. We return to that distinction in Section 11.

3 Model and method

Each electron m carries a density u_m^2 supported on a domain fixed by a weight w_m, with $\int w_m u_m^2 = n_m$ the domain's occupation. Domains compete for territory, so the partition has free internal

boundaries. The energy is

$$E = \sum_m \left[\tfrac{1}{2} \int w_m |\nabla u_m|^2 - \int w_m K u_m^2 \right] + E_{ee} + \sum_{a<b} \frac{Z_a Z_b}{R_{ab}}, \qquad K(\mathbf{r}) = \sum_a \frac{Z_a}{r_{\mathrm{soft},a}}, \tag{2}$$

with every term weighted by w, electron–electron repulsion counting each pair once by the $i \neq j$ rule, and $r_{\mathrm{soft}}^2 = |\mathbf{r} - \mathbf{R}_a|^2 + (\sigma h)^2$ a kernel softening on the grid of spacing h. Calculations use a 100^3 to 160^3 grid, boxes of 12–24 a_0, bare kernels ($r_c = 0$) throughout, and 120 000 relaxation steps with early termination disabled.

3.1 The force must be the derivative of the discretised energy

The force on a nucleus in a model of this kind is the classical electrostatic force of the charge distribution, the content of the electrostatic force theorem [4, 5]; in a model built from charge densities it is definitional rather than derived. The protocol is only as good as the force, and applying it immediately exposed that the force in use was not a well-defined quantity. It was computed as $Z_a \nabla V_H$ at the nucleus, finite differenced and averaged over a sphere of radius a to suppress grid noise. Holding H_2 at a fixed separation $R = 1.68\,a_0$ and varying *only* that radius gives

a (bohr)	0.24	0.48	0.72	0.96
F (E_h/a_0)	−0.05019	−0.03227	−0.00624	+0.02706

There is no plateau and the *sign* changes: the computed force reports the smoothing radius. Two causes compound. The potential gradient is unresolved at the nuclear cusp on this mesh, and the averaging bias is $O(a^2 \nabla\rho)$ with $\nabla\rho$ largest exactly there.

The remedy is to differentiate the discretised energy analytically rather than to sample its gradient numerically. From $V_{eK} = -\sum_a Z_a \int \rho / r_{\mathrm{soft},a}$,

$$\mathbf{F}_a \;=\; -\frac{\partial V_{eK}}{\partial \mathbf{R}_a} \;=\; Z_a \int \rho(\mathbf{r})\, \frac{\mathbf{r} - \mathbf{R}_a}{r_{\mathrm{soft},a}^3}\, dV, \tag{3}$$

a volume integral with no averaging radius, no evaluation at the cusp, and—because it differentiates the same softened kernel from which the energy is built—agreement with $-dE/dR$ by construction, up to the term arising from the partition boundary moving with the nuclei—the analogue here of the basis-set derivative term of Pulay [6]. Validation on H_2, with force and energy taken from the same four calculations, places the zero of (3) at 1.84–1.89 a_0 against an energy minimum at 1.963 a_0, agreeing to 4–6%; the sphere-averaged force disagreed by 0.42 a_0 and had the wrong sign at $R = 1.68$.

4 Two implementations, and what each can reach

The measurements above were made with a GPU solver (`mol_fast`) that carries one domain per entry and caps at sixteen domains. That cap is the binding constraint on molecular size, and its meaning depends on the construction: with one domain per atom carrying Z_x electrons it permits sixteen atoms, which covers benzene, glycine, ethanol and comparable small organics; with the one-electron domains of the underlying construction it permits sixteen electrons, which covers CH_4 and C_2H_2 but not benzene.

The scope of the pseudo-kernel core model itself, measured on two-centre systems, is reported separately [3]. A second solver (`molecule.js`) implements the same model with one global field partitioned by labels, enforcing true non-overlap, and admits two thousand domains. It has been run on alpha-helices and on chignolin, so molecular size is not the obstacle there. Its force, however, was the sphere-averaged potential gradient of Section 3.1, with an averaging radius of $3\,a_0$—larger than most bonds, so that the sphere around one nucleus encloses its neighbour. We have ported (3) to it, with two differences forced by its conventions. Its kernel is *clamped* rather than softened, $r_{\mathrm{eff}} = \max(r, r_{\mathrm{clamp}})$, so the potential is exactly flat inside the clamp and those cells contribute identically zero to the force rather than a softened value. And the integral reads the density directly, so cells excluded by the core treatment carry no density and are omitted automatically.

The practical consequence is a gap that has only just been closed: until this port, the solver that scales past sixteen domains could not be validated by the present protocol, while the solver with a trustworthy force could not reach molecules of interest. Both the accuracy of the ported force and the behaviour of the protocol at protein scale remain to be established; the cost scaling is discussed in Section 5.

5 Cost

A volume integral over the grid is more expensive than a point gradient, and it is worth stating by how much. The point gradient visits $N_{\mathrm{atoms}} \times \frac{4}{3}\pi R^3 \approx 270\,N$ cells for a four-cell sphere. The volume form visits $N_{\mathrm{atoms}} \times S^3$, about $10^6 N$ on a 101^3 grid: some four thousand times more.

The relevant comparison, however, is with what the solver already does. One volume-force evaluation costs about the same as one update sweep of the electronic field, and *less* than one evaluation of the potential, which requires an elliptic solve of several sweeps per electron. Forces are moreover not evaluated every step: once per animation frame in one implementation, every ten electronic steps in the other, giving overheads of well under one percent and of order ten percent of the update cost respectively—the latter adjustable, since nuclei move slowly compared with the electronic relaxation.

At protein scale the $N \times S^3$ product would warrant a cutoff radius. It is worth noting that such a cutoff behaves oppositely to the smoothing radius it replaces: truncating a convergent integral has an error that *falls* with radius, and distant neutral groups contribute as dipole/r^3 rather than charge/r^2, so a plateau in the cutoff is attainable and meaningful. The smoothing radius of the point gradient admitted no plateau because its bias *grew* as a^2. The same word denotes a convergent truncation in one case and a divergent bias in the other.

6 Three checks, and which one applies

Every calculation carries three internal checks. They are not interchangeable, and the choice between them is dictated by symmetry—a point we found easy to get wrong.

Kernel–kernel energy. $V_{KK} = \sum_{a<b} Z_a Z_b / R_{ab}$ is exact arithmetic and confirms the geometry *actually built*, independently of any physics. Every configuration reported below matched its predicted value; this is how fractional kernel placement, per-bond spacing and coincident kernels were verified.

Total force as a convergence criterion. The sum of nuclear forces is *not* automatically zero. It equals the net force the electron cloud exerts on the nuclei; only when the electronic state

is stationary does the cloud exert no net force, and then $\sum_a \mathbf{F}_a = 0$ follows by Newton's third law. The residual is therefore a direct measure of electronic non-convergence—and a more reliable one than any energy-based criterion, since a relaxation can sit on a long plateau with the energy changing imperceptibly while the state is far from settled.

But symmetry cancels it. In a symmetric molecule, mirror or rotational symmetry forces $\sum_a \mathbf{F}_a = 0$ regardless of accuracy. A small residual in CO_2 (we measure 0.06% of the largest force) establishes only that the algorithm preserves symmetry. The sum carries information only in directions symmetry does not protect: every direction for a heteronuclear molecule, the C_3 axis of NH_3, the C_2 axis of water.

Per-atom zeros. Where symmetry is complete, the useful check is instead a single atom's force. An atom at a centre of symmetry—the carbon of CO_2 or of CH_4—must feel exactly nothing. Unlike the sum, this cannot be satisfied by cancellation among atoms, so a nonzero value is unambiguously error and its magnitude is the per-atom floor.

7 Results

Table 1 collects the configurations. Forces are quoted with + meaning outward, so a positive entry indicates the bond wants to lengthen. Where a configuration fails its own check we report the failure rather than a bond error.

system	Z_x	force	ΔR	$\lvert\sum F\rvert/\lvert F\rvert_{\max}$	status
H_2	1,1	+0.1383	+30%	14.7%	marginal
N_2	3,3	−0.2505	−8.2%	3.7%	good
O_2	2,2	−0.1682	−9.7%	11.0%	fair
O_2	3,3	−0.2463	−14.3%	8.4%	fair
CO	4,2	—		38.5%	fails
CO_2	2,4,2	−0.1632	−7.2%	0.06%*	good
HCN	1,4,3	C≡N −7.8%, H–C +3.5%		5.5%	good
HCN	1,2,3	—		88.7%	fails
H_2O	2,1,1	—		140%	fails
H_2O	3,1,1	—		126%	fails
H_3 lin.	1,1,1	+0.0420	—	34.9%	fails
H_4 chain	1,1,1,1	+0.0316	—	7.4–16%	partial
C_2H_2	1,4,4,1	outward	—	25%	fails
C_2H_2	1,2,2,1	outward	—	28%	fails
C_2N_2	3,4,4,3	N inward, C outward		3.2%	good
NH_3	3,1,1,1	—		141% (z)	fails
NH_3 planar	3,1,1,1	—		70% (x)	fails
CH_4	**4,1,1,1,1**	0.036 outward	—	**0.6%**	**best**
CH_4	2,1,1,1,1	—		40%	fails

Table 1: Forces at measured geometries. *Symmetry-enforced; carries no accuracy information (Section 6). Configurations in which two different elements receive the same Z_x are degenerate—the model cannot distinguish them—and are omitted.

7.1 H_4 against an exact reference

One configuration can be scored against an exact method rather than experiment. Full configuration interaction in a minimal basis [9] places the equilibrium of the equally spaced H_4 chain at $a = 1.8\,a_0$, and the chain has one electron per site with no pseudopotential, so no core parameter is involved. At that geometry all four RealQM forces point outward—the model would expand a chain the exact method calls stationary—while over-binding by 14–18% per atom when each method is referred to its own atom. Over-bound and too long is the pattern H_2 shows at +30%.

8 Kernel depth decides convergence

The clearest results hold geometry, symmetry, mesh and every other input fixed and change only the central kernel's charge:

	shallow	deep	factor
CH_4 (identical T_d geometry)	C+2: 40%	C+4: 0.6%	60
HCN (identical geometry)	C+2: 88.7%	C+4: 5.5%	16

A deep kernel holds its domain and fixes its boundary against the neighbours; a shallow one leaves the partition undetermined. The direction is worth emphasising: C+4 is the *less* resolved kernel, at $Zh = 0.48$ against a working guideline of $Zh \lesssim 0.1$, so depth helps convergence more than coarse resolution hurts it.

Three alternative explanations were tested and excluded.

- **Not hydrogen content.** CH_4 carries four hydrogens and is the cleanest polyatomic measured; HCN carries one and reaches 5.5%; CO at 4,2 carries none and fails at 38.5%. Hydrogen is neither necessary nor sufficient for failure.
- **Not the box.** The net force on water is 0.216, 0.215, 0.214 in boxes of 12, 18 and 24 a_0 with the mesh held fixed—0.9% across a doubling of the box. The state is box-converged and the residual is a property of it. The control calculation on H_2 changes *sign* with box and with mesh, marking its own error as stochastic.
- **Not off-axis placement.** Tetrahedral ligands are maximally off-axis and CH_4 is nearly exact, because the cube diagonals are a symmetry of the grid.

8.1 Cubic-grid anisotropy

A second, smaller effect is visible in planar NH_3, which has D_{3h} symmetry and must therefore give $\sum_a \mathbf{F}_a = 0$ in every direction; it gives 70% of the largest force along x. Resolving by ligand: the one lying on the grid axis has radial force 0.0802, the two off-axis ones 0.0545 and 0.0544. The off-axis pair agrees with itself to 0.2% and differs from the on-axis ligand by 47%.

A cubic grid has fourfold symmetry in its planes and threefold symmetry about its body diagonals. Molecular geometries that share those symmetries—linear, and tetrahedral—are represented evenly; threefold-in-plane and bent geometries are not. This accounts for NH_3 and H_2O being the two worst configurations measured and CH_4 among the best, and it is a property of the discretisation rather than of the model.

9 Valence conventions, measured

Z_x is a modelling choice, and the protocol makes it testable. A depth ladder at fixed mesh, scored both by the symmetry-breaking error and by the bond error:

	O+2	O+3	O+6 (single domain)	N+3	N+5
pair asymmetry	19.5%	**3.7**%	1.4%	6.0%	6.1%
ΔR	−6.4%	**−7.1**%	+59%	−8.1%	−13.7%

O+3 is the better oxygen: a fivefold gain in convergence over O+2 at the same bond error. N+3 stands, full valence costing accuracy while buying no convergence. Most instructive is O+6 as a single six-electron domain, which converges best of all and yields a bond 59% too long—it converges accurately to the wrong answer, which is exactly the failure mode the protocol is designed to make visible. So a rule of "use the deepest Z_x" is false beyond the valence structure the atomic model selects; depth helps only where the domain was genuinely under-bound.

It is worth noting that O+3 is one of the two valence shells in the atomic configuration 2+3+3 obtained independently from atomic calculations [2], a convergence of atomic and molecular evidence on the same number. Carbon admits no corresponding reduction: C+2 would freeze into the core the $2s$ pair that hybridises into the bonds, cannot form four bonds, and—since Z_x is the only label distinguishing elements—makes carbon identical to O+2.

10 Kernel dynamics

Releasing the kernels and letting them move under the force is both the sharpest test of the force and the capability a simulator is finally for. It is also where the distinction between a force and a sampled gradient stops being academic: a force that is not the gradient of anything conserves no energy, so in sustained dynamics it pumps or drains the system indefinitely.

Released from either side of the expected minimum, H_2 relaxes to

start	relaxes to	residual force	direction
1.40	1.8502	$[-0.0178, +0.0057]$	expanded
2.40	2.0932	$[+0.0128, -0.0168]$	contracted

Both move the correct way and bracket $[1.850, 2.093]$, whose midpoint 1.972 sits 0.5% from the energy minimum at 1.963 and inside the force's own zero crossing of 1.84–1.89 measured at fixed geometry. Three independent routes—force zero, energy minimum, and dynamical relaxation from both directions—therefore agree to a few percent, where before the force was repaired the first two disagreed by 27% and the force pointed the wrong way at $R = 1.68$.

The bracket is not a failure to converge but a measured precision. The residual energy drift between runs is $\sim 0.011\,E_h$; through the curvature $k = 0.370\,E_h/a_0^2$ that corresponds to a displacement $\sqrt{2(0.011)/0.370} = 0.24\,a_0$, and the bracket is $0.243\,a_0$ wide. The two relaxations converged to exactly the precision the unconverged electronic state permits, and tightening the geometry requires tightening the electronics, not the dynamics.

Two cautions follow, both measured rather than anticipated. The relaxed states carry energies about $0.011\,E_h$ *above* the fixed-geometry scan at comparable separation, the electronic field lagging behind moving nuclei in the familiar way; so relaxation yields trustworthy geometry and untrustworthy energies, and the two should be taken from different calculations. And at three centres the

electronic relaxation does not settle at all, oscillating by $\sim 0.03\,E_h$ out to 3×10^5 steps, so kernel dynamics there would track a wandering density. Dynamics is at present a two-centre capability.

10.1 What the old force does in dynamics

The same relaxation under the sphere-averaged gradient settles at 1.5203 with residual forces $[+0.01821, +0.00020]$—both of the same sign, so the molecule carries a net translational force and drifts bodily, which no isolated system may do. In sustained dynamics that is a spurious acceleration of the whole system, and in a thermostatted simulation it would appear as heating indistinguishable from physical dissipation.

It also lands *closer to experiment*: 1.520 against the measured 1.401, where the repaired force gives 1.97. This is worth stating plainly because it inverts the naive test. The repaired force agrees with the model's own energy minimum to 0.5%; it reports faithfully what the model says, and what the model says is wrong by 40%. The old force was wrong *about the model* in a way that partly compensated for the model's own error. Validating a force against experiment alone would have preferred the defective one; only the internal checks—the lone atom feeling exactly zero, the vanishing of $\sum_a \mathbf{F}_a$, and agreement with $-dE/dR$—reject it. Fixing the instrument made agreement with measurement worse and moved the error from the instrument to the model, which is where it can be addressed.

11 Relation to $-dE/dR$

In a force-law model the force is definitional and the geometry is where it vanishes, so agreement with the energy gradient is not required for the model to be well posed. It remains worth checking for three reasons. A force that is not a gradient conserves no energy, which in dynamics produces spurious heating indistinguishable from physical dissipation. It is a strong and free check on implementation: it is what exposed the sign-flipping force of Section 3.1, which was otherwise unremarkable—smooth, antisymmetric, and crossing zero at a plausible separation. And once the implementation is correct, the residual disagreement measures the contribution from the partition boundary moving with the nuclei, a term specific to a free-boundary model and not otherwise accessible.

12 Bearing on previously recorded three-centre failures

Earlier work in this programme recorded that every three-centre system attempted descended monotonically inward through its experimental bond, across nuclear charges, with and without a core, for odd and closed-shell systems alike. Measured by force at the observed geometry instead, CO_2 gives -7.2% and HCN -7.8%: ordinary bounded errors of the same size as the diatomics, not a collapse. Those scans were assembled from an energy relaxation that we find oscillates by $\sim 0.03\,E_h$ out to 3×10^5 steps, read through a stopping criterion that compares energies between wall-clock samples and cannot detect it. The recorded failures are therefore consistent with artefacts of the scan rather than properties of the model, and the systems should be re-measured before any mechanism is proposed for them.

13 Limitations

The bond errors reported here are not competitive with established electronic structure methods, and are not offered as such; they measure where a parameter-free real-space construction currently stands. Several specific limitations bound the numbers.

All calculations use bare kernels, so the deepest, $Z = 4$, sits at $Zh = 0.48$, far above the resolution guideline. No ΔR quoted here carries a mesh ladder, and the percentages should be read as indicative; the directions, the error bars and the convergence ranking are robust to this, since they rest on comparisons at fixed mesh.

The convergence problem is characterised but not solved. We can now measure electronic non-convergence, attribute it principally to shallow kernels, and exclude three rival explanations; we cannot yet remove it. Configurations containing shallow kernels or threefold and bent geometries remain unusable, which includes water and ammonia—a serious limitation for any application to hydrogen-bonded systems.

Measured geometries, force constants and ionisation energies are taken from the standard compilations [7, 8] and are recorded as unverified in the source files; they are the one external input to an otherwise parameter-free calculation.

Finally, the construction pins the electron count of each domain, so charge transfer between unlike atoms is representable only through the shape of each domain and not through its occupation. For homonuclear systems this costs nothing; for a heteronuclear molecule it constrains precisely the asymmetry that defines it.

14 Conclusion

Evaluating forces at observed geometries gives a validation protocol for a force-law molecular model that requires no reference calculation, no energy scan and no minimum-finding, and that converts directly to a bond-length error. Applied to RealQM it first exposed that the force in use was not a physical quantity, and then isolated kernel depth as the dominant source of error by controlled comparison, excluding hydrogen content, box size and off-axis placement by direct test. It produced measured verdicts on valence conventions, identified an anisotropy of the cubic grid affecting threefold and bent geometries, and indicated that a class of previously recorded failures is likely an artefact of unconverged scans.

The protocol's most useful property is not any individual number but its behaviour on failure. Of the configurations examined, those that are unreliable announce it through their own internal checks—the total force exceeds the signal, or an atom at a centre of symmetry feels a force it cannot feel—rather than returning a plausible wrong geometry. An energy scan returns a number for every configuration, reliable or not. For a model under development, a method that reports its own failures is worth more than one that reports only results.

References

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[9] Full configuration interaction in a minimal (STO-6G) basis for equally spaced hydrogen chains H_N, generated for this work: equilibrium at $a = 1.8\,a_0$ with $E = -2.19038$ ($N = 4$) and -3.26674 ($N = 6$), each referred to its own atom $E(\mathrm{H}) = -0.471039$. The hydrogen chain is a standard benchmark for many-electron methods, having one electron per site and no pseudopotential, so no core parameter enters the comparison.

[10] Public code, the molecule simulator of Section 1, and an interactive gallery of all experiments in the present work: `https://claes542.github.io/RealMolecule/gallery.html` (source repository: `https://github.com/Claes542/RealMolecule`).